1 Null Space and Column Space

Given $A \in M^{m\times n}(F)$, we have seen that we can define $T_A \in \mathscr{L}(M^{n\times 1}(F), M^{m\times 1}(F))$ by multiplying $x \in M^{n\times 1}(F)$ by the matrix A. We will now define 4 subspaces, 2 of $M^{n\times 1}(F)$ and 2 of $M^{m\times 1}(F)$, associated to A called the *Fundamental Matrix Spaces*.

Definition 1.1 (Null space of A, Nullity of A)**.** The *null space of* A, denoted by Nul A, is the kernel of the linear map T_A. That is,

$$\begin{aligned}\text{Nul } A := \ker T_A &= \{x \in M^{n\times 1}(F) : T_A x = 0\} \\ &= \{x \in M^{n\times 1}(F) : Ax = 0\},\end{aligned}$$

which is the solution set of the homogeneous linear system $Ax = 0$. The dimension of Nul A is called the *nullity of* A.

Definition 1.2 (Column space of A)**.** We define the *column space of* A, denoted by Col A, to be the image of the linear map T_A. Denoting the ith column of the matrix A by $A_{\bullet,i}$, we have

$$\begin{aligned}\text{Col } A := \text{im } T_A &= \{T_A x : x \in M^{n\times 1}(F)\} \\ &= \{Ax \in M^{n\times 1}(F)\} \\ &= \left\{\sum_{i=1}^{n} x_i A_{\bullet,i} : x_i \in F\right\} \\ &= \text{span}\{A_{\bullet,1}, \ldots, A_{\bullet,n}\}.\end{aligned}$$

That is, the column space of A is the span of the column vectors of the matrix A. The dimension of the column space of A is called the *rank of* A.

By the Fundamental Theorem of Linear Maps,

$$\begin{aligned}\dim M^{n\times 1}(F) &= \dim \ker T_A + \dim \text{im } T_A \\ n &= \dim \text{Nul } A + \dim \text{Col } A \\ n &= \text{nullity } A + \text{rank } A.\end{aligned}$$

Suppose first that A is in RREF.

- Since each non-pivot column is a linear combination of the pivot columns to the left of it, by the Plus/Minus Lemma the non-pivot columns can be removed from the set of column vectors without changing Col A. Since the set of pivot columns is linearly independent (by definition of RREF), this implies that rank A = # of pivot columns.
- Each non-pivot column gives rise to a free variable. As we have seen, the dimension of the solution set of the system $Ax = 0$ is equal to the number of free variables, so nullity A = # of non-pivot columns. Thus, the equation $n = \text{nullity } A + \text{rank } A$ simply expresses the fact that the total number of columns of A is the sum of the pivot and non-pivot columns.

If A is not in RREF, then it is still true that the pivot columns of A are a basis of Col A. This is because elementary row operations do not change the linear dependence relations among the columns of A. [1]

[1] A linear dependence relationship among the columns of A is a non-trivial solution of the equation $Ax = 0$. If A and B are row-equivalent, then $Ax = 0$ and $Bx = 0$ have exactly the same solution set.

These observations give rise to an algorithm for producing explicit bases of Nul A and Col A:

(1) Row-reduce A to its RREF, R.

(2) Use R to identify the pivot columns of A, which are a basis of Col A.

(3) Use R to find a parametric description of the solution set of $Ax = 0$. The vectors spanning the solution set are a basis of Nul A.

Example 1.3. Consider the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$

The reduced row echelon form of A is

$$R = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The first and third columns of A are pivot columns, so a basis of Col A is given by

$$\left\{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \right\}.$$

Since x_2, x_4, x_5 are free variables, so we set

$$\begin{aligned} x_2 &= t_1 \\ x_4 &= t_2 \\ x_5 &= t_3. \end{aligned}$$

A vector $x \in F^5$ is therefore a solution to this system of equations if it is of the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2t_1 + t_2 - 3t_3 \\ t_1 \\ -2t_2 + 2t_3 \\ t_2 \\ t_3 \end{bmatrix} = t_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t_3 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix},$$

therefore a basis of Nul A is given by

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Exercise 1.4. Find explicit bases of the null space and column space of the matrix

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}.$$

2 Row Space and Left Null Space

Definition 2.1 (Transpose of a matrix)**.** If A is an $m \times n$ matrix, then A^T denotes the *transpose* of A, which is the $n \times m$ matrix obtained by interchanging rows and columns of A.

Example 2.2. If $A = \begin{bmatrix} 2 & 1 & -3 \\ 5 & 6 & 7 \end{bmatrix}$, then $A^T = \begin{bmatrix} 2 & 5 \\ 1 & 6 \\ -3 & 7 \end{bmatrix}$.

Definition 2.3 (Row space of A)**.** We define the *row space of A* to be the span of the row vectors of the matrix A. Denoting the ith row of A by $A_{i,\bullet}$,

$$\text{Row } A := \text{span}\{A_{1,\bullet}, \cdots, A_{m,\bullet}\}.$$

Note that $\text{Row } A = \text{Col } A^T$.

Suppose A is in RREF.

- By definition of RREF, the first row is nonzero, and the remaining nonzero rows are not linear combination of the previous ones. Thus, these are linearly independent and therefore are a basis of Row A.
- Since each nonzero row contains a pivot, this shows that $\dim \text{Row } A = \#$ of pivots $= \dim \text{Col } A =$ rank A.

Suppose now that A is not in RREF. Let R be the RREF of A.

- Since elementary row operations are invertible, they don't change the row space, so we conclude that the nonzero rows of R are a basis of Row A.
- Since elementary row operations don't change pivot positions, we again conclude that $\dim \text{Row } A = \dim \text{Col } A = \text{rank } A$.

Example 2.4. Recall from Example 1.3 the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix},$$

which has reduced row echelon form

$$R = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

A basis for Row A is given by the non-zero rows of R:

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \\ -2 \end{bmatrix} \right\}$$

Definition 2.5 (Left null space)**.** The *left null space of A* is defined as the null space of A^T. That is, Nul (A^T) is the solution set of the homogeneous system $A^T y = 0$, or, equivalently, the system $y^T A = 0^T$ (which is why it is called the left null space).

We have just seen that we can find a basis of Row $A = \text{Col } A^T$ without row-reducing A^T. We can also find a basis of Nul A^T without having to also row-reduce A^T. We can view $y^T A$ as a linear combination of the rows of A:

$$y^T A = \sum_{i=1}^{n} y_i A_{i,\bullet}$$

The condition that y is in the left null space of A is that this linear combination is the zero (row) vector. Suppose we form the augmented matrix $[A \mid b]$ before row-reducing to RREF. There will be $n - \text{rank } A$ zero rows, which is exactly the dimension of the left null space. Each zero rows of A will be of the form $[0\,0\cdots 0|\alpha_1 b_1 + \cdots + \alpha_m b_m]$ for some scalars $\alpha_1, \ldots, \alpha_m$. For each zero row, we have found a linear combination of the rows of A (by the sequence of elementary row operations reducing A to RREF) which has produced the zero row. The vector of coefficients $\begin{bmatrix}\alpha_1 & \cdots & \alpha_m\end{bmatrix}$ is therefore an element of the left null space of A. It follows from the definition of RREF that these vectors are linearly independent, and hence a basis of the left null space of A.

Example 2.6. We will find explicit bases for all 4 fundamental matrix spaces of the matrix $A = \begin{bmatrix}1 & 2\\ 2 & 3\\ 2 & 4\\ 2 & 5\end{bmatrix}$. Add an augmented column and row-reduce to RREF:

$$A = \begin{bmatrix}1 & 2 & b_1\\ 2 & 3 & b_2\\ 2 & 4 & b_3\\ 2 & 5 & b_4\end{bmatrix} \mapsto R = \begin{bmatrix}1 & 0 & 5b_1 - 2b_4\\ 0 & 1 & -2b_1 + b_4\\ 0 & 0 & -2b_1 + b_3\\ 0 & 0 & -4b_1 + b_2 + b_4\end{bmatrix}$$

A is a 4×2 matrix. The rank of A is 2, so the nullity is zero since these must add up to 2.
A^T is a 2×4 matrix. Since rank $A^T = \text{rank } A = 2$, the left null space has dimension 2 since these must add up to 4.

- A basis of Col A is given by the pivot columns of A: $\left\{\begin{bmatrix}1\\2\\2\\2\end{bmatrix}, \begin{bmatrix}2\\3\\4\\5\end{bmatrix}\right\}$
- Nul $A = \left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}$, so a basis of Nul A is given by $\emptyset$.
- A basis of Row A is given by the nonzero rows of R: $\{\begin{bmatrix}1 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\end{bmatrix}\}$
- The entries in the augmented column of the zero rows of R are $-2b_1 + b_3$ and $-4b_1 + b_2 + b_4$. The coefficient vectors give a basis of the left null space of A: $\{\begin{bmatrix}-2 & 0 & 1 & 0\end{bmatrix}, \begin{bmatrix}-4 & 1 & 0 & 1\end{bmatrix}\}$

Exercise 2.7. Find explicit bases of all 4 fundamental matrix spaces of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$